# CREDIT CARD

# FINANCIAL PERFORMANCE ANALYSIS

SQL + Power BI | End-to-End Data Analytics Project

**Dataset Source: Kaggle | 10,000+ Records | FY 2023**

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**Tools**: PostgreSQL | Power BI | Power Query | DAX

## Project Overview

| | |
|---|---|
| **Project Title** | Credit Card Financial Performance Analysis |
| **Dataset Size** | 2 CSV Files with 10,108 records (initial) + 185 records added (Week 53 refresh) |
| **Time Period** | January 2023 to December 2023 (53 Weeks) |
| **Feature** | **Automated data refresh pipeline simulating weekly live reporting.** |

| | |
|---|---|
| **SQL Concepts Used** | JOINs, GROUP BY, ORDER BY, CASE WHEN, CTEs, Window Functions, NULLIF, ROUND, UNION ALL, Subqueries |
| **DAX Concepts Used** | SUM, AVERAGE, DIVIDE, CALCULATE, DATEADD, IF, FORMAT, ABS, COUNTROWS, FILTER, ALL, ALLSELECTED, CALENDAR |

# Business Requirements

## Revenue & Financial Performance

- What is the total revenue, interest income, and transaction volume for the reporting period?
- How does revenue break down across card categories?
- Which quarters and weeks show peak and low revenue performance?
- What is the week-over-week revenue change, is the business growing or declining?

## Customer Segmentation & Behaviour

- How does revenue differ between male and female customers?
- Which age groups, income segments, and occupations drive the most revenue?
- How do education level and dependent count correlate with spending patterns?
- What expenditure types & payment methods dominate card usage?

## Risk & Operational Metrics

- How are customers utilizing their credit limits across card categories?

## Geographic Performance

- Which states generate the highest revenue and transaction volume?

## Data Pipeline & Automation

- Dashboard should automatically reflect new weekly data without manual intervention.

# Dataset Overview

**Two CSV files used as source data**, imported into PostgreSQL and joined on the client_num key to enable cross-table analysis, then **to check the pipeline** added cc_add.csv and cust_add.csv

## cc_detail: Credit Card Transaction Table

| | |
|---|---|
| **Records** | 10,108 (initial) + 185 added via cc_add.csv |

## cust_detail: Customer Demographics Table

| | |
|---|---|
| **Records** | 10,108 unique customers + 185 added via cust_add.csv |

# SQL Queries & Results

## Database Setup & Importing Data

```sql
-- Create cc_detail table

CREATE TABLE cc_detail (
    Client_Num INT,
    Card_Category VARCHAR(20),
    Annual_Fees INT,
    Activation_30_Days INT,
    Customer_Acq_Cost INT,
    Week_Start_Date DATE,
    Week_Num VARCHAR(20),
    Qtr VARCHAR(10),
    current_year INT,
    Credit_Limit DECIMAL(10,2),
    Total_Revolving_Bal INT,
    Total_Trans_Amt INT,
    Total_Trans_Ct INT,
    Avg_Utilization_Ratio DECIMAL(10,3),
    Use_Chip VARCHAR(10),
    Exp_Type VARCHAR(50),
    Interest_Earned DECIMAL(10,3),
    Delinquent_Acc VARCHAR(5)
);
```

```sql
-- Create cust_detail table

CREATE TABLE cust_detail (
    Client_Num INT,
    Customer_Age INT,
    Gender VARCHAR(5),
    Dependent_Count INT,
    Education_Level VARCHAR(50),
    Marital_Status VARCHAR(20),
    State_cd VARCHAR(50),
    Zipcode VARCHAR(20),
    Car_Owner VARCHAR(5),
    House_Owner VARCHAR(5),
    Personal_Loan VARCHAR(5),
    Contact VARCHAR(50),
    Customer_Job VARCHAR(50),
    Income INT,
    Cust_Satisfaction_Score INT
);
```

```sql
COPY cc_detail
FROM 'D:\DA Prep\Projects\SQL + Power bi\Credit Card Finance\credit_card.csv'
DELIMITER ','
CSV HEADER ;

COPY cust_detail
FROM 'D:\DA Prep\Projects\SQL + Power bi\Credit Card Finance\customer.csv'
DELIMITER ','
CSV HEADER ;
```

## Data Exploration (Initial Row count was - 10108)

```sql
SELECT * FROM cc_detail ;
```

Data Output Messages Notifications

| | client_num integer | card_category character varying (20) | annual_fees integer | activation_30_days integer | customer_acq_cost integer | week_start_date date | week_num character varying (20) | qtr character |
|---|---|---|---|---|---|---|---|---|
| 1 | 708082083 | Blue | 200 | 0 | 87 | 2023-01-01 | Week-1 | Q1 |
| 2 | 708083283 | Blue | 445 | 1 | 108 | 2023-01-01 | Week-1 | Q1 |
| 3 | 708084558 | Blue | 140 | 0 | 106 | 2023-01-01 | Week-1 | Q1 |
| 4 | 708085458 | Blue | 250 | 1 | 150 | 2023-01-01 | Week-1 | Q1 |
| 5 | 708086958 | Blue | 320 | 1 | 106 | 2023-01-01 | Week-1 | Q1 |
| 6 | 708095133 | Blue | 100 | 0 | 94 | 2023-01-01 | Week-1 | Q1 |

```sql
SELECT * FROM cust_detail;
```

Data Output Messages Notifications

| | client_num integer | customer_age integer | gender character varying (5) | dependent_count integer | education_level character varying (50) | marital_status character varying (20) | state_cd character varyin |
|---|---|---|---|---|---|---|---|
| 1 | 708082083 | 24 | F | 1 | Uneducated | Single | FL |
| 2 | 708083283 | 62 | F | 0 | Unknown | Married | NJ |
| 3 | 708084558 | 32 | F | 1 | Unknown | Married | NJ |
| 4 | 708085458 | 38 | M | 2 | Uneducated | Single | NY |
| 5 | 708086958 | 48 | M | 4 | Graduate | Single | TX |
| 6 | 708095133 | 33 | F | 1 | High School | Single | NY |

```sql
-- Total customers, total transactions
SELECT count(DISTINCT client_num) FROM cc_detail ;
SELECT count(DISTINCT client_num) FROM cust_detail ;

-- Date range
SELECT min(week_start_date) as start_date, max(week_start_date) as end_date
FROM cc_detail;
```

Data Output Messages Notifications

| | start_date date | end_date date |
|---|---|---|
| 1 | 2023-01-01 | 2023-12-31 |

```sql
-- How many unique card categories exist?
SELECT DISTINCT card_category FROM cc_detail;
```

Data Output Messages Notifications

| | card_category character varying (20) |
|---|---|
| 1 | Silver |
| 2 | Gold |
| 3 | Platinum |
| 4 | Blue |

```sql
-- Check for nulls or anomalies
SELECT * FROM cc_detail WHERE client_num IS NULL;
SELECT * FROM cust_detail WHERE client_num IS NULL;
SELECT * FROM cc_detail WHERE card_category IS NULL;
```

Data Output Messages Notifications

| client_num integer | card_category character varying (20) | annual_fees integer | activation_30_days integer | custome integer |
|---|---|---|---|---|

## Revenue Analysis

### KPIs:

```sql
SELECT
    SUM(annual_fees + total_trans_amt + interest_earned) AS total_revenue,
    SUM(interest_earned) AS total_interest,
    SUM(total_trans_amt) AS transaction_amount,
    COUNT(DISTINCT client_num) AS total_customers
FROM cc_detail;
```

ata Output Messages Notifications

| total_revenue numeric | total_interest numeric | transaction_amount bigint | total_customers bigint |
|---|---|---|---|
| 56517010.810 | 7982479.810 | 45533021 | 10293 |

### Revenue, Average Utilization, Transaction count, Interest earned and Annual Fees by Card category

```sql
SELECT
    card_category,
    SUM(annual_fees + total_trans_amt + interest_earned) AS total_revenue,
    ROUND(AVG(avg_utilization_ratio)::NUMERIC,3) AS avg_utilization,
    SUM(total_trans_ct) AS transactions,
    SUM(interest_earned) AS interest_earned,
    SUM(annual_fees) AS annual_fees
FROM cc_detail
GROUP BY card_category
ORDER BY total_revenue DESC;
```

a Output Messages Notifications

| card_category character varying (20) | total_revenue numeric | avg_utilization numeric | transactions bigint | interest_earned numeric | annual_fees bigint |
|---|---|---|---|---|---|
| Blue | 47188611.620 | 0.292 | 591791 | 6614172.620 | 2733690 |
| Silver | 5659108.980 | 0.083 | 49172 | 821922.980 | 189590 |
| Gold | 2533682.160 | 0.125 | 18781 | 384755.160 | 57565 |
| Platinum | 1135608.050 | 0.151 | 7490 | 161629.050 | 20665 |

## Revenue & Transaction count by Quarters

```sql
SELECT
    qtr,
    SUM(annual_fees + total_trans_amt + interest_earned) AS total_revenue,
    COUNT(*) AS transaction_count
FROM cc_detail
GROUP BY qtr
ORDER BY qtr;
```

a Output Messages Notifications

| qtr character varying (10) | total_revenue numeric | transaction_count bigint |
|---|---|---|
| Q1 | 13964362.770 | 2534 |
| Q2 | 13820566.720 | 2535 |
| Q3 | 14235479.690 | 2535 |
| Q4 | 14496601.630 | 2689 |

## Weekly Revenue Trend

```sql
SELECT
    week_num,
    week_start_date,
    SUM(annual_fees + total_trans_amt + interest_earned) AS weekly_revenue
FROM cc_detail
GROUP BY week_num, week_start_date
ORDER BY week_start_date;
```

Data Output Messages Notifications

| | week_num character varying (20) | week_start_date date | weekly_revenue numeric |
|---|---|---|---|
| 1 | Week-1 | 2023-01-01 | 1035629.320 |
| 2 | Week-2 | 2023-01-08 | 1053088.810 |
| 3 | Week-3 | 2023-01-15 | 1148249.800 |
| 4 | Week-4 | 2023-01-22 | 1071919.270 |

# Customer Segmentation Analysis

## Revenue by Gender, Age Group, Income Group & Education Level

```sql
SELECT
    cu.gender,
    SUM(cr.annual_fees + cr.total_trans_amt + cr.interest_earned) AS total_revenue
FROM cc_detail cr
JOIN cust_detail cu ON cr.client_num = cu.client_num
GROUP BY cu.gender
ORDER BY total_revenue DESC;
```

Output Messages Notifications

| gender character varying (5) | total_revenue numeric |
|---|---|
| M | 30929733.660 |
| F | 25587277.150 |

```sql
SELECT
    CASE
        WHEN cu.customer_age <= 30 THEN '20-30'
        WHEN cu.customer_age <= 40 THEN '31-40'
        WHEN cu.customer_age <= 50 THEN '40-50'
        WHEN cu.customer_age <= 60 THEN '50-60'
        ELSE '60+'
    END AS age_group,
    SUM(cr.annual_fees + cr.total_trans_amt + cr.interest_earned) AS total_revenue
FROM cc_detail cr
JOIN cust_detail cu ON cr.client_num = cu.client_num
GROUP BY age_group
ORDER BY total_revenue DESC;
```

Output Messages Notifications

| age_group text | total_revenue numeric |
|---|---|
| 40-50 | 25583024.240 |
| 50-60 | 16288922.060 |
| 31-40 | 11587587.410 |
| 60+ | 1695425.380 |
| 20-30 | 1362051.720 |

```sql
SELECT
    CASE
        WHEN cu.income <= 35000 THEN 'Low'
        WHEN cu.income <= 70000 THEN 'Mid'
        ELSE 'High'
    END AS income_group,
    SUM(cr.annual_fees + cr.total_trans_amt + cr.interest_earned) AS total_revenue
FROM cc_detail cr
JOIN cust_detail cu ON cr.client_num = cu.client_num
GROUP BY income_group
ORDER BY total_revenue DESC;
```

Output Messages Notifications

| income_group text | total_revenue numeric |
|---|---|
| High | 29841026.070 |
| Mid | 16115653.190 |
| Low | 10560331.550 |

```sql
SELECT
    cu.education_level,
    SUM(cr.annual_fees + cr.total_trans_amt + cr.interest_earned) AS total_revenue
FROM cc_detail cr
JOIN cust_detail cu ON cr.client_num = cu.client_num
GROUP BY cu.education_level
ORDER BY total_revenue DESC;
```

Output Messages Notifications

| education_level character varying (50) | total_revenue numeric |
|---|---|
| Graduate | 22785679.930 |
| High School | 11434012.270 |
| Unknown | 8666468.050 |
| Uneducated | 8312915.130 |
| Post-Graduate | 2819671.010 |
| Doctorate | 2498264.420 |

# Revenue & Customer Satisfaction Score by Occupation

```sql
SELECT
    cu.customer_job as occupation,
    SUM(cr.annual_fees + cr.total_trans_amt + cr.interest_earned) AS total_revenue,
    SUM(cu.income) AS total_income,
    ROUND(AVG(cu.cust_satisfaction_score)::NUMERIC, 2) AS avg_css_score
FROM cc_detail cr
JOIN cust_detail cu ON cr.client_num = cu.client_num
GROUP BY cu.customer_job
ORDER BY total_revenue DESC;
```

Output Messages Notifications

| occupation character varying (50) | total_revenue numeric | total_income bigint | avg_css_score numeric |
|---|---|---|---|
| Businessman | 17697472.010 | 190350431 | 3.23 |
| White-collar | 10283123.920 | 105618475 | 3.10 |
| Selfemployeed | 8542826.400 | 77659931 | 3.22 |
| Govt | 8335533.840 | 90834727 | 3.22 |
| Blue-collar | 7040606.420 | 73516911 | 3.18 |
| Retirees | 4617448.220 | 49619308 | 3.17 |

# Transaction Behaviour Analysis

## Revenue by Expenditure Type & Transaction Mode

```sql
SELECT
    exp_type,
    SUM(annual_fees + total_trans_amt + interest_earned) AS total_revenue
FROM cc_detail
GROUP BY exp_type
ORDER BY total_revenue DESC;
```

Output Messages Notifications

| exp_type character varying (50) | total_revenue numeric |
|---|---|
| Bills | 14001372.300 |
| Entertainment | 9781401.680 |
| Fuel | 9587104.420 |
| Grocery | 8734990.550 |
| Food | 8375136.950 |
| Travel | 6037004.910 |

```sql
SELECT
    use_chip,
    SUM(annual_fees + total_trans_amt + interest_earned) AS total_revenue,
    ROUND(
        SUM(annual_fees + total_trans_amt + interest_earned) * 100.0 /
        SUM(SUM(annual_fees + total_trans_amt + interest_earned)) OVER(), 2
    ) AS percentage
FROM cc_detail
GROUP BY use_chip
ORDER BY total_revenue DESC;
```

Output Messages Notifications

| use_chip character varying (10) | total_revenue numeric | percentage numeric |
|---|---|---|
| Swipe | 35662565.260 | 63.10 |
| Chip | 17338110.670 | 30.68 |
| Online | 3516334.880 | 6.22 |

## Delinquency Rate Accounts with Overall %

```sql
SELECT
    COUNT(*) AS total_accounts,
    SUM(CASE WHEN delinquent_acc = '1' THEN 1 ELSE 0 END) AS delinquent_accounts,
    ROUND(
        SUM(CASE WHEN delinquent_acc = '1' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2
    ) AS delinquency_rate_pct
FROM cc_detail;
```

Output Messages Notifications

| total_accounts bigint | delinquent_accounts bigint | delinquency_rate_pct numeric |
|---|---|---|
| 10293 | 624 | 6.06 |

## Geographic Analysis: Top 5 states by Revenue

```sql
SELECT
    cu.state_cd,
    SUM(cr.annual_fees + cr.total_trans_amt + cr.interest_earned) AS total_revenue
FROM cc_detail cr
JOIN cust_detail cu ON cr.client_num = cu.client_num
GROUP BY cu.state_cd
ORDER BY total_revenue DESC
LIMIT 5;
```

Output Messages Notifications

| state_cd character varying (50) | total_revenue numeric |
|---|---|
| TX | 13015112.410 |
| NY | 12971862.620 |
| CA | 12901378.910 |
| FL | 9969060.330 |
| NJ | 4381264.040 |

# Week-over-Week Revenue Analysis

This is the core analysis of the Weekly Status Report using a CTE and LAG window function to calculate revenue change between consecutive weeks.

```sql
WITH weekly_revenue AS (
    SELECT
        week_num,
        week_start_date,
        SUM(annual_fees + total_trans_amt + interest_earned) AS current_week_revenue
    FROM cc_detail
    GROUP BY week_num, week_start_date
),
wow_comparison AS (
    SELECT
        week_num,
        week_start_date,
        current_week_revenue,
        LAG(current_week_revenue) OVER (ORDER BY week_start_date) AS prev_week_revenue
    FROM weekly_revenue
)
SELECT
    week_num,
    week_start_date,
    current_week_revenue,
    prev_week_revenue,
    (current_week_revenue - prev_week_revenue) AS wow_change,
    ROUND(
        (current_week_revenue - prev_week_revenue) * 100.0 /
        NULLIF(prev_week_revenue, 0), 2
    ) AS wow_change_pct
FROM wow_comparison
ORDER BY week_start_date DESC;
```

ta Output Messages Notifications

| week_num character varying (20) | week_start_date date | current_week_revenue numeric | prev_week_revenue numeric | wow_change numeric | wow_change_pct numeric |
|---|---|---|---|---|---|
| Week-53 | 2023-12-31 | 1201600.580 | 933134.430 | 268466.150 | 28.77 |
| Week-52 | 2023-12-24 | 933134.430 | 1070439.100 | -137304.670 | -12.83 |

| | week_num character varying (20) | week_start_date date | current_week_revenue numeric | prev_week_revenue numeric | wow_change numeric | wow_change_pct numeric |
|---|---|---|---|---|---|---|
| 1 | Week-53 | 2023-12-31 | 1201600.580 | 933134.430 | 268466.150 | 28.77 |
| 2 | Week-52 | 2023-12-24 | 933134.430 | 1070439.100 | -137304.670 | -12.83 |
| 3 | Week-51 | 2023-12-17 | 1070439.100 | 1026549.110 | 43889.990 | 4.28 |
| 4 | Week-50 | 2023-12-10 | 1026549.110 | 980152.370 | 46396.740 | 4.73 |
| 5 | Week-49 | 2023-12-03 | 980152.370 | 1008776.600 | -28624.230 | -2.84 |
| 6 | Week-48 | 2023-11-26 | 1008776.600 | 1047120.330 | -38343.730 | -3.66 |
| 7 | Week-47 | 2023-11-19 | 1047120.330 | 1078915.240 | -31794.910 | -2.95 |
| 8 | Week-46 | 2023-11-12 | 1078915.240 | 1094926.590 | -16011.350 | -1.46 |
| 9 | Week-45 | 2023-11-05 | 1094926.590 | 1063063.370 | 31863.220 | 3.00 |

## Data Refresh Validation

To simulate a real-world weekly data pipeline, 185 new records (Week 53) were added to both tables using the COPY command. Power BI was then refreshed to verify the dashboard updated automatically.

### Copying and appending newly updated data into both tables.

(Initial Row count was - 10108)

```sql
-- Adding new credit card transaction data
COPY cc_detail
FROM 'D:\DA Prep\Projects\SQL + Power bi\Credit Card Finance\cc_add.csv'
DELIMITER ','
CSV HEADER ;

-- Adding new customer data
COPY cust_detail
FROM 'D:\DA Prep\Projects\SQL + Power bi\Credit Card Finance\cust_add.csv'
DELIMITER ','
CSV HEADER ;

-- Row counts after data addition
SELECT 'cc_detail' AS table_name, COUNT(*) AS total_rows FROM cc_detail
UNION ALL
SELECT 'cust_detail', COUNT(*) FROM cust_detail;
```

Output Messages Notifications

| table_name text | total_rows bigint |
|---|---|
| cc_detail | 10293 |
| cust_detail | 10293 |

### Checking the new week data

```sql
SELECT
    week_num,
    week_start_date,
    COUNT(*) AS records_added,
    SUM(annual_fees + total_trans_amt + interest_earned) AS revenue
FROM cc_detail
WHERE week_start_date = (SELECT MAX(week_start_date) FROM cc_detail)
GROUP BY week_num, week_start_date;
```

Output Messages Notifications

| week_num character varying (20) | week_start_date date | records_added bigint | revenue numeric |
|---|---|---|---|
| Week-53 | 2023-12-31 | 185 | 1201600.580 |

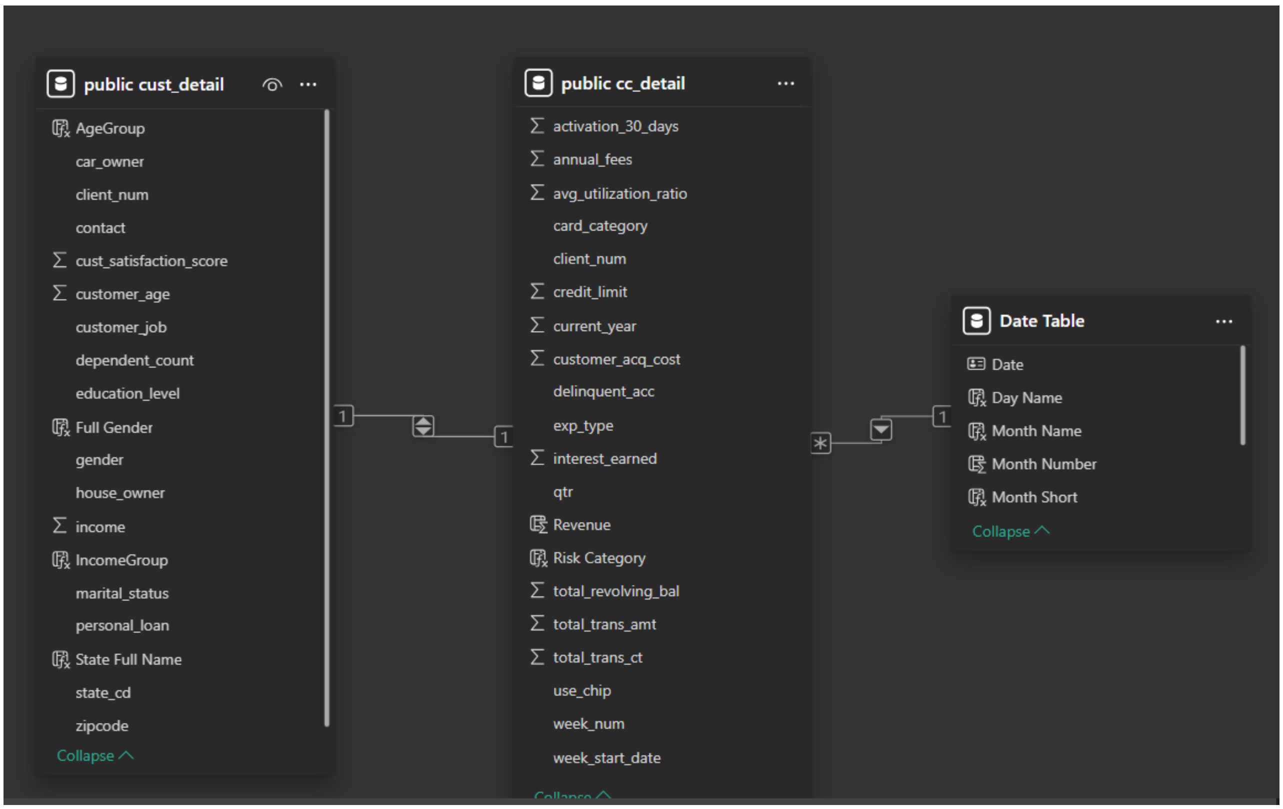
**After refreshing Power BI,**

**Total Revenue updated from 55M to 57M, Transaction Count from 656K to 667K, and Week 53 appeared in the WoW table with a 28.8% growth <u>confirming the automated refresh pipeline works correctly.</u>**

# Power BI Dashboard Overview

Two interactive dashboards were built in Power BI, connected to PostgreSQL. The dashboards use a custom date table.

## Data Model



## Dashboard 1: Credit Card Transaction Report

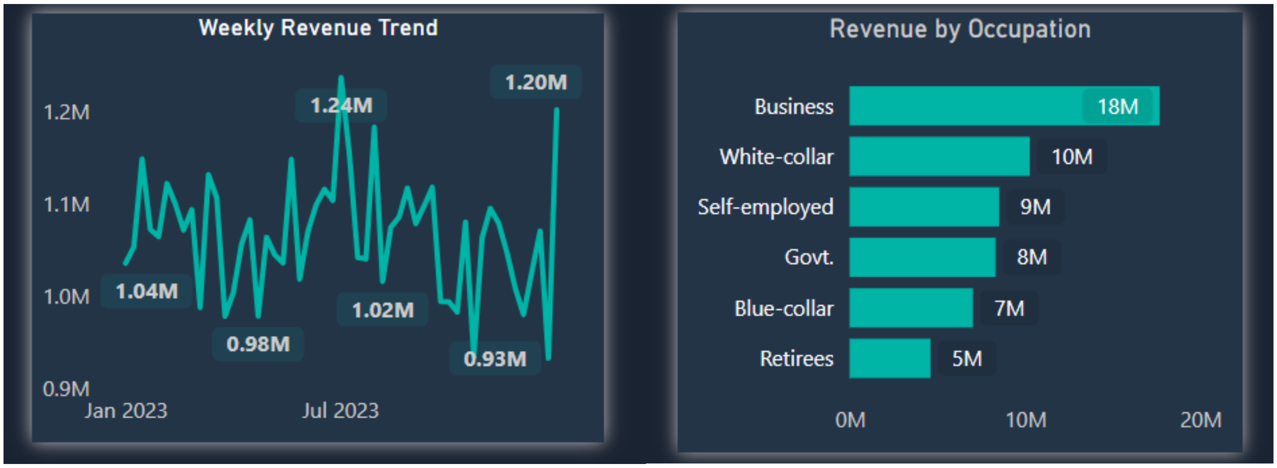
Focuses on transaction-level performance, weekly trends, and revenue by expenditure and card type.


## Dashboard 2 — Credit Card Customer Intelligence Report

Focuses on customer demographics, satisfaction, income segmentation, and geographic performance.

## Tooltip Pages

Two report page tooltips were built to provide drill-down context on hover:




# Key Insights & Business Findings

**1**

**Blue Card Revenue Concentration:**
Blue cardholders generate 83% of total revenue despite being the entry-level, lowest-fee card tier. This extreme revenue concentration in a single segment represents a significant retention risk. If Blue customers churn, the business loses the majority of its revenue base.
<u>Strategy recommendation:</u> Implement Blue cardholder loyalty programs and proactive upgrade campaigns to Silver/Gold.

**2**

**Premium Card Satisfaction Paradox:**
Platinum cardholders report the lowest customer satisfaction score (2.72) despite paying the highest annual fees and being the premium product tier. Gold customers (3.05) report better satisfaction than Platinum. This signals that premium value proposition is not meeting customer expectations.
<u>The business should audit Platinum benefits vs competitor offerings and conduct customer exit interviews.</u>

**3**

**Self-Employed Credit Reliance**
The Income vs. Spending scatter analysis reveals that Self-employed customers show disproportionately high credit card spending relative to their average income, higher credit reliance than their income level would suggest. This represents both a revenue opportunity (high spenders) and a portfolio risk (potential over-leveraging).
<u>Delinquency monitoring for this segment is recommended.</u>

4

### Digital Payment Adoption Gap as 'Swipe' Dominates

63% of transactions are conducted via physical card swipe the least secure payment method. Only 6.2% of transactions use online payment channels. This low digital adoption rate represents a security risk and a missed opportunity for digital engagement.

Targeted campaigns promoting chip and online payment usage could reduce fraud risk while improving customer digital experience.

5

### Q3 Peak Performance

Q3 generated the highest revenue (14.2M) and transaction count across all four quarters.

Q4 shows a notable decline to 13.3M despite the holiday season suggesting the customer base may not include high retail spenders.

Understanding the drivers of Q3 outperformance could inform targeted Q4 revenue recovery campaigns.

6

### Graduate Education Segment Drives Spending

Graduate-level customers generate the highest revenue nearly double High School customers and significantly more than Post-Graduate customers. Education level is a stronger predictor of credit card spending than income group alone.

7

### High Income Group: Female Customers Outspend

In the High income segment, female customers generate 23M vs male customers at 7M, a 3x difference. This gender spending gap in the premium income bracket is a significant insight for targeted marketing. High-income female customers represent an underserved and high-value acquisition target for premium card tiers.

8

### Week 53 Refresh: Pipeline Validation Confirmed

After appending 185 new records (Week 53) to the PostgreSQL database and refreshing Power BI, Total Revenue updated from 55M to 57M, Transaction Count from 656K to 667K, and Week 53 appeared at the top of the WoW table with 28.8% growth. This confirms the end-to-end automated data pipeline functions correctly, new data flows from SQL to Power BI without any manual dashboard updates required.